



MINING DEALER BEST PRACTICE

777D/F/G Final Drive Replacement Stand

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The service team of one of the Borusan Makina Kazakhstan's projects has made a stand that is used with the forklift to safely replace final drives of 777D/F/G trucks at the site.

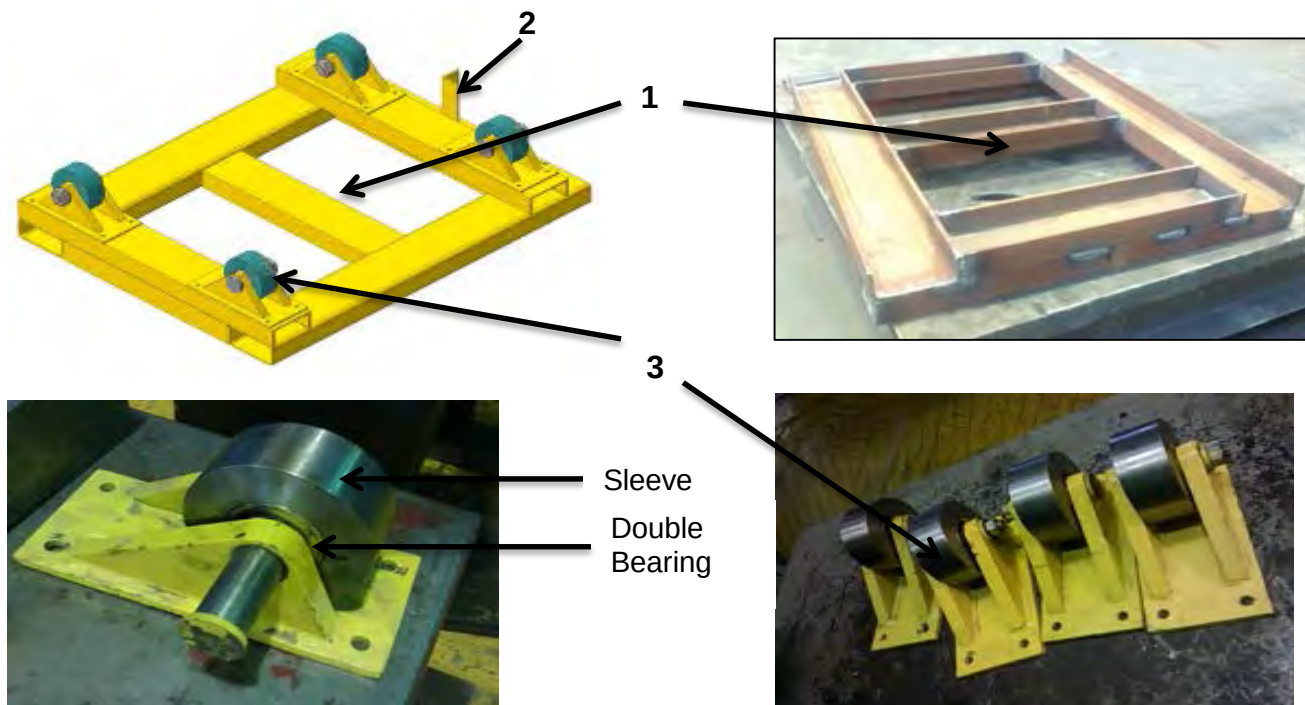
Previously for replacing the final drive, only the forklift was used. The final drive was delivered by the forklift to the location of the final drive in the truck. Then in order to match the holes in the machine and the holes in the final drive, the technician had to turn the final drive with the help of the crowbar (the weight of the final drive more than 1.5 ton). It could cause the final drive to drop from the forklift and might hurt the technician or damage the component itself.

This stand helps the service team to replace final drives safely and quickly.

2.0 Best Practice Description

This stand consists of two main parts: 1.The stand itself and; 2.The arc-shaped band for the planetary part of the final drive.

The **stand** has the **Foundation** (1) from the Channels 120 and 140, and 200 for the forks of the big size (universal) with **Stopper** (2) that prevent the final drive falling from the stand and 4 x **Rollers** with double bearings that are pressed into the sleeves by the 5-ton capacity press and bolted to the stand (3). The sleeves of the rollers allow expanding the span with the contact surface. These kind of rollers provide necessary movement of the final drive during its installation into the truck.



The stand and its components

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The **arc-shaped band** is installed into the planetary side of the final drive. It has an arc surface, two walls and the retainers from both sides. The band has two types of holes. One is for bolts with washers; the others are made to reduce the weight of the band. This band provides necessary movement of the planetary side of the final drive during its installation into the truck.



The arc-shaped band

3.0 Implementation Steps

The 777D/F/G Final Drive Replacement Stand is simple in use.

1. Use the crane to put the final drive on the stand.
2. While the final drive is hanging by the crane, bolt the arc-shaped band to the planetary side of the final drive.
3. Put the final drive on the stand.
4. Use the forklift to transport the final drive to its location in the truck.
5. Install the final drive to the truck by slightly turn the final drive in order to match the holes of the final drive to the holes in the truck.
6. Bolt the final drive to the truck.
7. Then remove the forks of the forklift and unbolt the arc-shaped band.



BEFORE



AFTER

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4.0 Benefits

- Safe, easy and quick replacement of final drives of 777D/F/G trucks in the site.
- Efficient utilization of man hours.

5.0 Resources Required

- Welding Shop.
- Material and parts required are indicated in the drawings attached.

6.0 Supporting Attachments / References

The detail and assembly drawings of the 777D/F/G Final Drive Replacement Stand are attached.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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